



The influence of venture capital syndicate size on venture performance

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ABSTRACT

Venture capital (VC) syndicates pool diverse resources from their members to accomplish the common goal of nurturing new ventures for a successful exit. Although the size of syndicate is a fundamental attribute impacting performance, the influence of syndicate size is less understood in prior studies with mixed findings. To address the gap, we suggest that there is an inverted U relationship between a syndicate size and venture performance. As the number of partners in a VC syndicate increases, a syndicate can provide more heterogeneous resources that can help its portfolio company succeed, but coordination costs increase as well. We thus predict that the net effect combining these two countervailing effects yields an inverse-U relationship between syndicate size and performance. We further examine two boundary conditions under which the nonlinear relationship is likely to manifest. Analyzing 407 investment syndicates formed by 1,106 VC firms for new ventures in the U.S. information and communications technology sector between 1990 and 2006, we find that the relationship between syndicate size and performance is an inverse-U shape. We further find that geographic distance among syndicate partners flattens the inverse-U curve, whereas a strong reputation of the lead VC firms shifts the inverse-U curve to the right.

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Introduction

New ventures are resource-constrained and depend on external investors to exploit entrepreneurial opportunities (Stinchcombe 1965). In particular, venture capitalists (VCs) play an important role in the development and performance of new ventures by providing not only financial capital, but also managerial resources such as industry expertise and business acumen, professionalization of managers, and contacts for potential customers and collaborators (Gompers and Lerner 2004; Hellmann and Puri 2002; Hsu 2006; Huang and Knight 2017; Sahlman 1990). In most cases, a syndicate of VC firms, rather than a single VC firm, make venture investments as VC firms seek to reduce their risk exposure by pooling funds in multiple ventures with other VC firms to achieve successful outcomes (Lerner 1994; Sorenson and Stuart 2001). VC firms also use their syndication networks as

an important vehicle to access external knowledge and deal flows (De Clercq and Dimov 2008; Sorenson and Stuart 2008).

A number of studies have examined the formation of syndication and partner selection. For example, VC firms are likely to co-invest with those who are similar in terms of the level of experience (Lerner 1994; Du 2016). Although VC firms usually prefer local firms as syndicate partners, they also co-invest with those who are geographically distant or have different investment profiles when the target company's industry is fashionable or involves less uncertainty (Sorenson and Stuart 2008). Prior ties based on co-investment experiences and group dynamics also influence the formation of syndication and partner selection (Zhang, Gupta, and Hallen 2017; Zhang and Guler 2020). Another stream of research has examined the performance of syndication (e.g., De Clercq and Dimov 2008; Sharifzadeh and Walz 2012). This line of research has shown that investment in industries in which a VC firm has more expertise and investment with familiar partner firms in the syndicate tend to enhance the investment performance (De Clercq and Dimov 2008). Although VC firms are likely to co-invest with others who are similar to them, they benefit more from syndicating with partners that are different from them in terms of the success of their portfolio firms (Du 2016). Moreover, co-investing with foreign VC firms also enhances the chance of a successful exit of portfolio firms (Chahine, Saade, and Goergen 2019).

Among important characteristics of VC syndicates, syndicate size is a fundamental attribute affecting its performance because the resources available on a team and the dynamics of collaboration depend on primarily how many members are on it (Hambrick and D'Aveni 1992; Zhang, Gupta, and Hallen 2017). However, the effect of syndicate size on venture performance is not understood well (Jääskeläinen 2012). Prior studies on the topic have given only secondary attention or treated it as an alternative explanation to be controlled for. Both theoretical arguments and empirical findings from prior studies are also equivocal. Some research finds that larger syndicates are advantageous by highlighting the collaborative benefits based on the pooled resources and capabilities from multiple members in the syndicate (Giot and Schwenbacher 2007), whereas others find the opposite by focusing on the internal agency problems and coordination challenges associated with a large syndicate (Dimov and De Clercq 2006; Falconieri, Filatotchev, and Tastan 2019). These mixed findings indicate that the relationship between syndicate size and performance may be nonlinear, calling for proper theorization on the underlying mechanisms. Yet, the inquiry remains under-researched (c.f. De Clercq and Dimov 2008).

This study aims to provide a theoretical basis for the influence of syndicate size on venture performance with empirical evidence. Drawing on the literature on group (e.g., Goodman, Ravlin, and Argote 1986), we suggest that there is an inverted U relationship between a syndicate size and venture performance. As more VC firms join an investment syndicate, the magnitude and heterogeneity of resources that syndicate partners provide will increase and enhance the outcome of the portfolio companies. However, we expect this positive effect of resource heterogeneity to have a diminishing marginal productivity on venture performance due to the possibility of redundant resources and skillsets that additional syndicate members can bring. On the other hand, a large syndicate could incur greater coordination costs among multiple partners. Such costs tend to increase exponentially as the number of constituents involved in formulating venture strategy increases. Combining these two non-linear functions (Haans, Pieters, and He 2016), we

predict that the resulting relationship between VC syndicate size and venture performance is inverse-U shaped.

To further understand the net effect of benefits and costs associated with a large syndicate size, we consider how the curvilinear relationship is moderated by factors that influence the dynamics of VC syndicate partners. We first consider how geographic dispersion among syndicate partners influences the curvilinear relationship between syndicate size and performance. The moderating effect of geographical dispersion among syndicate members is not immediately clear. On the one hand, greater benefits from heterogeneous resources stemming from firms in different geographic regions may outweigh the increase in coordination costs from the additional partners, thereby shifting the inverse-U curve between syndicate size and performance upward. On the other hand, the additional coordination costs may increase more rapidly relative to the increase in benefits from heterogeneous resources offered by the geographically scattered partners, shifting the inverse-U curve downward. We thus offer two competing hypotheses on these two possibilities.

Moreover, a large group tends to put more pressure and demands on the group's ability to make timely decisions due to multiple perspectives. Thus, a lead VC firm's ability to create a consensus for smooth decision-making is critical. Because members are more likely to respect the opinion of a capable leader, syndicates with reputable lead VC firms may be able to accommodate a greater number of partners with a relatively slower increase in coordination costs, compared to other syndicates led by VC firms with relatively less reputation. At the same time, reputable lead VC firms are also more likely to bring other partners with more heterogeneous resources to the syndicate and help the portfolio companies succeed. This effect is likely to be particularly stronger when the syndicate size increases. Therefore, we expect that when a highly reputable VC firm leads a syndicate, the inverse-U curve will shift outward.

We test these predictions by analyzing 1,106 VC firms that invested in 407 new venture syndicates in the U.S. information and communications technology (ICT) sector. We find that the relationship between syndicate size and the likelihood of a new venture's successful exit event in terms of going public is indeed inverted-U shaped. We also find that geographic dispersion among partners makes the inverse-U curve flatter, whereas the reputation of the lead VC firm shifts the curve to the right.

Our study makes three contributions. First, we contribute to entrepreneurship research focusing on resource assembly through VC syndicates. Our results help entrepreneurs understand the implications of receiving funding from multiple investors in a large syndicate in the efforts to acquire external resources. Second, our study contributes to research on VC syndicate by providing nuanced understanding regarding the influence of syndicate size. By explicating the two latent mechanisms – collaboration benefits and coordination challenges – underlying the nonlinear relationship between syndicate size and venture performance, we reconcile the inconsistent findings in the relevant prior studies. We further identify the two boundary conditions under which the nonlinear relationship is likely to manifest. Third and more broadly, our study complements the research on multiparty collaboration (e.g., Heidl, Steensma, and Phelps 2014; Zhang, Gupta, and Hallen 2017) by assessing the benefits and costs of collaboration and their performance implication in the context of VC syndicate.

Theory and hypotheses

Venture capital syndicate

Syndicates are prevalent in VC investment activities (Gompers and Lerner 2004; Sorenson and Stuart 2001; Sahlman 1990). VC firms form syndicates for several reasons. First, syndication enables VC firms to better select the venture in which to invest (Cumming 2006; Sahlman 1990). Because investing in a new venture is highly risky, it is beneficial to co-invest with other VC firms that can provide a second opinion on its potential before making an investment. In doing so, VC firms can reduce the risk exposure to a particular venture by sharing commitment while investing in multiple ventures with finite capital. Second, syndication serves as a mechanism through which to pool resources, experience, and complementary skills from multiple partners that can help investees achieve better outcomes (Brander, Amit, and Antweiler 2002; Tian 2011). Third, syndication provides VC firms with greater opportunities to participate in future deals through the exchange of information among syndication partners (Manigart et al. 2006), including opportunities for investments in non-local ventures. Although VC firms typically prefer investing in local ventures for better monitoring and nurturing, they can also invest in non-local ventures by co-investing with other VC firms in which such ventures would be a local investment (Sorenson and Stuart 2001). Fourth, syndication can serve as a governance mechanism to curb opportunism by VC firms. Because VC firms often rely on syndication networks for future deal flows, a tainted reputation can jeopardize future opportunities (Axelroad 1984; Poppo, Zhou, and Ryu 2008; Zhang, Gupta, and Hallen 2017). As a result, VC firms would be less likely to behave opportunistically when they work with other VC firms that can oversee their behavior.

Despite these benefits, VC syndicates entail coordination challenges as multiple firms work together to garner expected benefits (Zhang, Gupta, and Hallen 2017). It is generally more difficult to encourage cooperation from multiple partners vis-a-vis dyadic relationships due to the higher anonymity of individual action, more diffused harm, and lack of controllability of partner behavior in a multiparty setting (Fox and Guyer 1977). There are also unique challenges in VC syndicates. First, VC firms have different risk propensities. Some carry a high-risk strategy with the hope of obtaining a “home run” with a few ventures, whereas others prefer a relatively low-risk strategy that focuses on “singles” (Norton and Tenenbaum 1993). VC firms may also differ in terms of the extent to which they get involved in venture development. Some may exercise a “laissez faire” governance style, whereas others are “close trackers” with hands-on management (MacMillan, Kulow, and Khoylian 1989). Relatedly, VC firms may also have different ideas regarding the growth strategy of a venture such as patenting, the level of product or service quality, market entry timing, or replacement of the top management team. Likewise, VC firms may prefer different exit strategies depending on, in part, their investment time horizons. Some VC firms prefer an IPO as an exit strategy, whereas others may prefer selling the venture to another firm as a way to quickly harvest investment. Depending on the strength of a VC firm’s prior experience, divisions may occur among syndicate partners that could cause conflict and prevent joint decision-making when a rapid decision is critical (George, Erikson, and Parhankangas 2016; Heidl, Steensma, and Phelps 2014). Moreover, a mismatch between power and status with venture equity ownership may result in dysfunctional syndicates,

particularly when a new venture receives funding from a large syndicate early in its life (Ma, Rhee, and Yang 2013). In sum, large VC syndicates produce two simultaneous yet countervailing forces on the performance of the syndicate, suggesting that the relationship is likely to be nonlinear. In particular, although both resource benefits and coordination costs are expected to increase with syndicate size in different ways, prior research has examined only linear relationships and the findings have been mixed at best (Dimov and De Clercq 2006; Falconieri, Filatotchev, and Tastan 2019; Giot and Schwienbacher 2007; Jääskeläinen 2012).

The relationship between syndicate size and venture performance

Several reasons suggest that large VC syndicates may be associated with enhanced performance of their portfolio companies. VC firms differ in terms of the amount of available capital funding for investment offerings and also possess different skills, experiences, network contacts, and knowledge sets based on the scope of investment experience and scope. Advantages that firms can bring may include organizing production, contacts for potential business partners and customers, or recruiting important human capital (Hochberg, Lindsey, and Westerfield 2015). A large syndicate could also increase the likelihood that a venture would secure needed capital in subsequent rounds (Zhang, Gupta, and Hallen 2017). These heterogeneous and complementary resources through syndication suggest that a larger syndicate is likely to exhibit enhanced performance for its portfolio company (Brander et al. 2002; Giot and Schwienbacher 2007; Tian 2011). Nevertheless, the benefits of a large syndicate associated with resource heterogeneity are likely to diminish in marginal productivity given that multiple partners are likely to have duplicate sets of resources and skillsets.

On the other hand, an increasing number of syndicate partners could also lead to greater coordination costs and impede the collaboration effort (Zhang, Gupta, and Hallen 2017). For instance, VC firms typically take a board seat when they invest in a new venture (Garg 2013; Park and Steensma 2014). When they are unable to get a board seat, they tend to ask for an observer seat. As the number of partners increases, the need for communication on decision-making increases; however, arranging an in-person meeting with a greater number of partners becomes difficult. Even if the partners manage to gather together in the boardroom, a large number of VC firms bringing different perspectives and ideas will make it difficult to create a consensus. Moreover, as the size of the syndicate increases, the probability of an uncooperative firm (“bad apple”) in the syndicate also arises (Fox and Guyer 1977). Consequently, the board is unlikely to be effective due to the greater possibility of misalignment of interests with a large syndicate size. Relatedly, no individual VC firm in a large syndicate would own enough equity to be dominant. This can create a decision vacuum, especially when a venture goes sideways (Feld 2009). Moreover, as the size of syndication grows, the number of dyadic relationships in the syndication partners increases exponentially, which leads to a greater likelihood of personality conflicts (Amason and Sapienza 1997; George, Erikson, and Parhankangas 2016). As a result, coordination costs are also likely to increase exponentially when more members join the syndicate. As stated in an online media, “VCs are like martinis: the first is good, the second one great, and the third is a headache.”¹

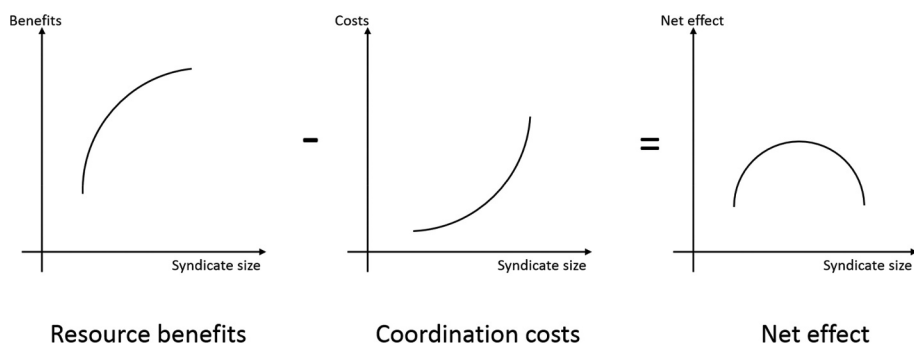


Figure 1. Mechanisms behind the inverted U-shaped relationship between syndicate size and performance.

In sum, we consider the influence of VC syndicate size on venture performance in terms of resource provision benefits and coordination costs. As shown in Figure 1, both resource benefits and costs increase as the number of VC firms in a syndication increases. However, the benefits of resource heterogeneity by a larger number of syndicate partners increase at a diminishing rate, whereas the coordination costs increase at an exponential manner when more partners join the syndicate. As a result, the net effect by subtracting the costs from the benefits generates an inverse-U relationship between syndicate size and venture performance. A larger VC syndicate is likely to increase the performance of its portfolio company up to a certain point, but after a certain threshold, the effect of a large number of syndicate partners will cause a decrease in performance. Therefore, we hypothesize:

Hypothesis 1. The number of partners in a VC syndicate will have an inverse-U relationship with venture performance such that a larger size syndicate will be associated with higher performance; however, beyond some level, a larger size syndicate will be associated with lower performance.

We have thus far argued that the benefits and costs of additional VC firms to a syndicate will generate an inverse-U relationship between syndicate size and portfolio company performance. In the next section, we seek to better understand the trade-off between the benefits and costs associated with a large syndicate size on venture performance by examining two contingencies: geographic dispersion and the reputation of the lead VC firm.

Geographic dispersion of VC firms and syndicate size

Despite the convenience of online meetings and advancement of telecommunications technologies, physical proximity provides distinctive advantages such as face-to-face interactions, exchange of rich information from each party and the contexts in which such interactions occur, rapid feedback, and ready access to mutual resources (Blau 1977; Chen et al. 2010). Because of these advantages, most social and economic exchanges frequently occur among people residing in the same or nearby locations (McPherson and Smith-Lovin 1987). As a result, VC firms generally prefer collaborating with firms operating

in close proximity. However, substantial variation also exists in terms of geographical distance among VC partners depending on firm age, experience, and other dimensions (Sorenson and Stuart 2001). We expect that geographic dispersion among partners may serve as an important contingency on the relationship between syndicate size and venture performance by influencing coordination costs as well as resource benefits for a syndicate with a large number of partners.

When geographic dispersion among VC firms is low (i.e., all firms are in close proximity), such firms tend to have redundant resources or overlapping contacts based on their regions. For example, when a new venture seeks a potential customer or talented individuals to join the management team, VC firms close to each other are likely to reach out to the same pool of resources. Thus, when a syndicate size increases but geographic dispersion among the VC firms is low, the increase in resource benefits tends to be smaller compared to when geographic dispersion is high. However, the obvious advantage of a syndicate consisting of VC firms located in close proximity is lower coordination costs associated with a large number of partner firms. In a large syndicate, the greater likelihood of misalignment of interests and personality conflicts can hinder collaboration when rapid decision-making is necessary. Face-to-face group discussions facilitated by geographic proximity helps partners overcome such issues by enabling the exchange of rich information regarding expertise and resources, and even in terms of individual managers' personality characteristics. When a syndicate encounters a problem in decision-making, in-person group discussions allow partner firms to gather information on the choices or options that others are willing to accept, which is useful in establishing group norms and introducing conformity pressures in favor of collective choices (Zeng and Chen 2003). Thus, geographic proximity among VC firms can reduce the coordination costs associated with a larger syndicate.

On the other hand, when geographic dispersion among VC firms is high (i.e., scattered in different geographic regions), partner firms could bring heterogeneous resources to the syndicate based on different regions. For example, when a new venture attempts to commercialize a product or develop a market, geographically dispersed VC firms might be helpful in finding business partners beyond the local region. Nonetheless, when VC firms in a syndicate are geographically dispersed, the syndicate is likely to experience increased coordination costs associated with a large number of partner firms. In such a case, not only are informal, ad-hoc meetings less likely to occur, it is also more difficult to arrange face-to-face meetings with all partners. Monitoring and advising young startups often entail a high level of uncertainty as well as complex issues in various aspects of the business. Although such startups could rely more on virtual communication channels, it is less desirable to discuss such complex issues due to the absence of mutual interactions and relative impatience of participants with regard to telephone conversations (Holstein and Gubrium 2003). Indeed, the quality of monitoring can be lower when geographical dispersion among syndicate members is large (Bernstein, Giroud, and Townsend 2016).

These arguments suggest that geographic dispersion among VC syndicate partner firms simultaneously influences the two underlying mechanisms that we focus on: the benefits from resource heterogeneity and coordination costs. Thus, how geographical dispersion shifts the inverted-U curve between syndicate size and venture performance will depend on which effect is more dominant – resource benefits or coordination costs.

However, no clear a priori theoretical reason suggests that one force is more dominant than the other. We thus develop two competing hypotheses.

Specifically, when the increased benefit of resource heterogeneity of adding a partner firm due to high geographic dispersion is larger than the associated increase in coordination costs, the inverted-U curve between syndicate size and venture performance will move upward, as shown in Figure 2(a). In contrast, when the increased benefit of resource heterogeneity of adding a partner firm due to high geographic dispersion is smaller than the associated increase in coordination costs, the inverted-U curve between syndicate size and venture performance will move downward, as shown in Figure 2(b). Thus, we propose:

Hypothesis 2a. Geographic dispersion of the VC syndicate will amplify the inverse-U shaped relationship between syndicate size and venture performance such that the peak (the inflection point) of the curve will be at a higher point when VC partners in a syndicate are more geographically dispersed.

Hypothesis 2b. Geographic dispersion of VC syndicate will attenuate the inverse-U shaped relationship between syndicate size and venture performance such that the peak (the inflection point) of the curve will be at a lower point when VC partners in a syndicate are more geographically dispersed.

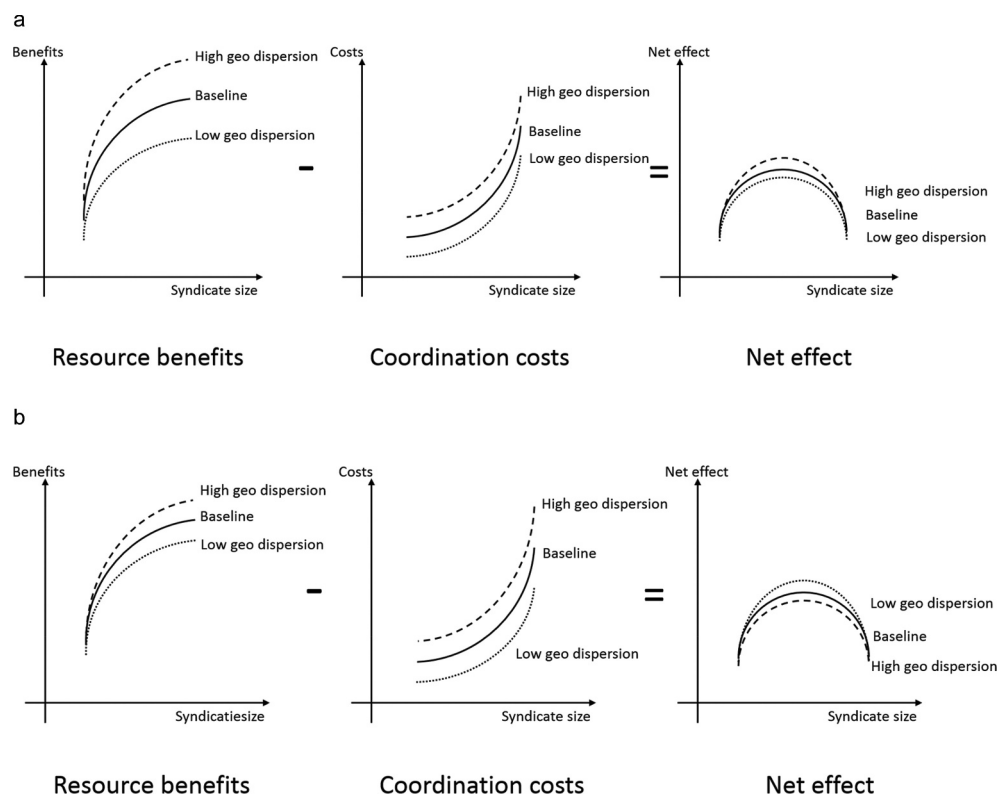


Figure 2. (a) The moderating effect of geographical dispersion: Resource benefits > coordination costs. (b) The moderating effect of geographical dispersion: Resource benefits < coordination costs.

Lead VC firm reputation

Another factor that may influence the benefits and costs of large syndicate size is the reputation of the lead investor. Syndication in the VC industry typically involves a lead investor that sets the valuation and other terms for investment contracts (Sorenson and Stuart 2008). Lead investors typically provide the largest amount of capital in each funding round and spend significantly more time managing and monitoring the focal venture as compared with other partners (Gorman and Sahlman 1989). Thus, such investors have a significant influence on the outcome of the venture. Lead investors are likely to be particularly important in larger syndicates, as there are often greater demands on the leadership role in coordinating activities to achieve the goal. In this regard, the reputation of the lead investor, in terms of prior experience in facilitating ventures to successful outcomes, is likely to determine the effect of syndicate size on venture performance by mitigating the coordination costs. When multiple actors attempt to achieve a common goal, highly reputable actors typically have a stronger influence based on their expertise in achieving the common goal. For instance, star scientists who possess the expertise in a scientific topic tend to exercise a disproportional amount of influence in determining the scientific agenda of an inventor team (Zucker and Darby 1996). Likewise, venture CEOs with greater scientific expertise (Park and Tzabbar 2016) or VC firms with superior track records of successful exits (Park and Steensma 2013) can have a stronger influence in determining venture strategy.

We propose that the inverted U-shape relationship between syndicate size and venture performance will shift outwards – i.e., toward the upper-right – when a syndicate is led by a highly reputable VC firm as compared to a less-reputable lead VC firm. On the one hand, reputable lead VC firms can bring syndication partners with heterogeneous resources that enhance the resource bundle available for their portfolio companies. Specifically, highly reputable VC firms are more likely to be aware of a startup's resource needs and mobilize resource pools possessed by other potential co-investors. They may also be more likely to be able to take advantage of these heterogeneous resources in a way that can help successful outcomes for their venture investees (Gompers and Lerner 2004). Although this ability could lead to the direct positive effect of lead VC reputation on the performances of portfolio companies, we suggest that such an effect would be particularly more pronounced when the syndicate size is large. Given their prior experiences in facilitating ventures to successful outcomes, reputable lead VC firms can anticipate resource needs and more effectively identify potential syndicate partners that possess valuable resources. Their invitations to those firms are also more likely to be accepted. Thus, the expertise and tacit knowledge of reputable lead VC firms would enable them to better assemble a syndicate with complementary resource bundles, and such expertise is likely to be more valuable when the syndicate size is large. Moreover, the balance between reputation and ability to influence the syndicate outcomes by a lead VC firm is likely to enhance the performance of portfolio companies (Ma, Rhee, and Yang 2013). Therefore, the resource benefits by a syndicate led by a reputable VC firm could increase more rapidly as compared with a syndicate led by a less reputable VC firm when the syndicate becomes large.

On the other hand, the presence of a highly reputable lead VC firm is likely to decrease the coordination costs incurred from a large syndicate, as such a firm would

exercise a stronger influence on other syndicate members. Specifically, a larger syndicate typically creates greater communication demands and a higher likelihood of conflict due to diverse perspectives and personalities. Also, as ownership becomes diluted with additional syndicate partners, no single owner has a dominant share, which could encourage free-riding behaviors and reduce individual incentives to contribute their resources. For example, De Clercq, Sapienza, and Zaheer (2008) found that larger syndicate size reduces each partner's voluntary involvement in venture development, such as providing professional contacts, helping with recruitment, and mentoring. All these factors make it difficult for the syndicate to engage in timely, joint decisions in dealing with the uncertain future of a new venture. In such cases, the opinions of highly reputable lead VC firms, in terms of prior experience in nurturing new ventures with successful exit events, have the credibility and legitimacy to overcome the dead-end situation. Reputable lead investors are also more likely to offer quality information from which the group can make better decisions or develop new ideas, thereby promoting universal collaboration. Thus, their opinions are likely to be well accepted by other syndicate partners.

In contrast, a syndicate led by a less reputable VC firm is likely to have greater challenges in forming a syndicate with valuable, heterogeneous resources and may not be able to orchestrate the available resources of different partners in a timely manner. This deficiency can be particularly problematic in a large syndicate. Such a VC firm might either lack the experience of facilitating successful outcomes or experienced failures in previous attempts. Therefore, the resource benefits by a syndicate led by a less reputable VC firm is not likely to increase as rapidly compared with another syndicate led by a more reputable VC firm when the syndicate becomes larger. However, the coordination costs of such syndicate led by a less reputable VC firm is likely to increase faster. When conflicts or difficult situations arise in which all partners feel unsure of the next step, the syndicate is likely to need more time for problem resolution or face a decision vacuum due to the lack of leadership compared to a syndicate with a more reputable lead VC firm. In such cases, some firms may even withdraw out of frustration. This would not only disrupt the relationships of remaining VC firms in the syndicate, but also create a concern that the venture may be unable to secure needed capital in the subsequent rounds (Zhelyazkov and Gulati 2016). These problems are likely to get worse in a large syndicate led by a less reputable VC firm.

Taken together, these arguments indicate that, compared to a large syndicate led by a less-reputable VC firm, a large syndicate led by a reputable VC firm is likely to enjoy greater resource benefits, whereas associated coordination costs are likely to decrease. Hence, the lead VC firm reputation will shift both the shape and turning point of the curve by enhancing resource benefits and alleviating coordination costs. Figure 3 illustrates our theoretical argument regarding the moderating effect of the lead VC firm's reputation. Therefore, we hypothesize:

Hypothesis 3. The lead VC firm's reputation will shift the inverse-U shaped relationship between syndicate size and venture performance up and outward such that the peak (the inflection point) of the inverse-U curve will shift to the upper-right when the lead VC firm is highly reputable.

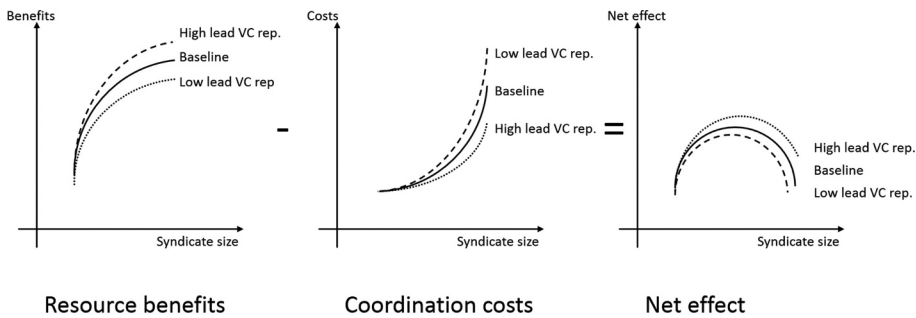


Figure 3. The moderating effect of lead VC reputation.

Methods

Data and sample

We examine the effect of VC syndicate size on the exit outcomes of its portfolio companies. Specifically, our sample consists of 501 syndicates formed by 1,124 VC firms that invested in U.S.-based startups in the information and communications technology (ICT) sector between 1990 and 2002. The startups funded by the syndicates were founded between 1990 and 2002 and received at least one round of VC financing between 1990 and 2006. We tracked the funding histories and exit events of these ventures through 2006. The ICT sector provides an appropriate setting for our study as it is one of the most heavily invested sectors by VCs over the past decade or so (Pitchbook 2020). Specifically, we sampled VC syndicates that invested in new ventures in the following industry subsectors: computer hardware (SIC code 357), communication equipment (SIC code 366), and telecommunications (SIC codes 481–484, 499). Of the 501 VC syndicates, 40 syndicates were dropped due to missing values in some of the control variables. We also included investments at the startup/seed, early, expansion, and later stages, but excluded other investments immediately prior to an exit event.² Thus, our final sample consisted of 407 investment syndicates formed by 1,106 VC firms during the period between 1990 and 2006. The unit of analysis is VC syndicate by year. Following prior research, we define a VC syndicate as “two or more venture capital firms taking an equity stake in an investment for a joint payoff, either in the same investment round or at different points in time (Lockett, Ucbasaran, and Butler 2006)

Our primary source of data was VentureXpert from Thomson One, which offers detailed data on various types of investor firms, their portfolio companies, and funding rounds including investment amount and date. We relied on the U.S. Patent and Trademark Office (USPTO) and patent citation data from the National Bureau of Economic Research’s (NBER) patent database (Hall, Jaffe, and Trajtenberg 2001) to construct some of our control variables. We also used COMPUTSTAT and the Securities Data Corporation (SDC) Platinum Alliances databases, and hand-collected data on founders’ prior employment history through an extensive online search using various sources such as company websites, *Businessweek*’s executive profiles and biography database, and the professional networking website, LinkedIn.

Measures

IPO

To examine venture performance, we measured whether a new venture invested by a focal syndicate (i.e., portfolio company) went public. We created a binary variable, *IPO*, which took a value of 1 if a new venture went public during the study period and 0 otherwise. Although IPOs are typically the most successful and preferred outcomes for VC firms, some research considers acquisition as another successful exit outcomes for start-ups (e.g., Graebner and Eisenhardt 2004; Ma, Rhee, and Yang 2013). Thus, we conduct robustness check with an alternative measure in which both IPOs and acquisitions are coded as a measure of successful venture performance and present the results in the robustness check section.

Number of VC firms

We measured syndicate size by counting the total number of VC firms investing in each funding round.

Geographic dispersion among syndicate partners

We calculated the geographic distance among syndication partners using a formula based on zip codes (Sorenson and Stuart 2001). Specifically, we calculated the dyadic distances of all pairs among syndication partners and a focal new venture, respectively. We then took the average distance across all pairs.

Lead VC firm reputation

We took several steps to create the lead VC firm's reputation. We first designated a focal new venture's lead VC firm for each round to be the one that invested with the largest amount. We then examined their investment portfolio to see the exit outcomes of the portfolio companies. For a given year t , we counted the total number of venture investees taken public prior to t . We took the log transformation to reduce skewness.

Control variables

We also included a host of factors that could affect a new venture's exit outcome (Fitza, Matusik, and Mosakowski 2009). Because investment relationships with industry incumbents have a significant effect on a new venture's exit outcome (Kim and Park 2017), we include *CVC investment relationship*, which is coded as 1 if a focal new venture has received funding from corporate investors and 0 otherwise. We assume that a relationship lasts for five years (Dushnitsky 2006). We also accounted for other relationships such as *Alliance relationships*, because collaboration with other firms may influence venture performance by providing endorsement benefits (Baum and Silverman 2004; Stuart, Hoang, and Hybels 1999). We assumed that this relationship lasts for three years (Ahuja 2000). We included *New venture innovation capability* by counting the forward citation count of a new venture's patents using a five-year moving window (Ahuja 2000; Stuart and Podolny 1996). We relied on the Harvard Dataverse (U.S. patent inventor database) (Lai et al. 2014) to create the measure. We included *Total investment amount* of a new venture by computing the cumulative amount invested in the firm until $t-1$ (Aggarwal and Hsu 2014; Alvarez-Garrido and Dushnitsky 2016). We took a log

transformation of this variable to reduce skewness. We also included several variables to account for the human capital of a new venture's founding team. We hand-collected individual career histories of 755 founders of the 407 new ventures. Specifically, we included *Prior industry experience*, which is coded as 1 if any of the entrepreneurs had any work experience in the ICT sector prior to founding the focal new venture. We also examined whether entrepreneur(s) had any prior entrepreneurial experiences. We included *Novice entrepreneur*, which is coded as 1 if none of the founder(s) of a new venture had company founding experiences prior to founding the focal new venture and 0 otherwise.

We accounted for a new venture's location by including two indicator variables, *California* (61% of our sample) and *Massachusetts* (9% of our sample). All other locations were used as a reference category. We also accounted for the industry differences by including binary indicators for each ICT subsector based on the three-digit SIC code. There may be macro-environmental factors that could influence a new venture's likelihood of going public. Thus, we included *Number of IPOs* by measuring the total number of new ventures that went public in a given year. We used data from the Ritter's database.³ During our 1990–2006 study period, the year 2000 was a “bubble” year for venture capital investment (Dushnitsky 2006). Therefore, we included a binary indicator to account for any difference in the macro-economic factors before and after the bubble period, which might have influenced new ventures' exit outcomes.

Results

Table 1 presents the descriptive statistics and correlations between all variables in our sample. We recognize that some factors may influence the number of VC firms in a syndicate as well as the likelihood of a new venture going public, thereby making the number of VC firms endogenous. For example, a new venture perceived as more capable and with more upside potential may receive funding from a greater number of VC firms and also more likely to achieve successful exit outcome. To the extent that these factors are observable, we account for them in our models with the control variables such as new venture innovation capability or entrepreneurs' prior industry experience; yet, there can still be some unobservable factors that influence both syndicate size and the likelihood of IPO. Thus, we used an empirical strategy to address the potential endogeneity. Because our dependent variable – i.e., *IPO*, is binary, whereas the endogenous explanatory variable *Number of VC firms* is continuous – we used an instrumental variable probit model with continuous endogenous explanatory variables. Formally, the model is

$$y_{1i}^* = y_{2i}\beta + x_{1i}\gamma + u_1$$

$$y_{2i} = x_{1i}\Pi_1 + x_{2i}\Pi_2 + v_i$$

where $i = 1, \dots, N$. y_1 is the dependent variable (i.e., *IPO*), and y_2 is $1 \times p$ vector of endogenous variable (i.e., number of VC firms). x_{1i} is a $1 \times k$ vector of exogenous variables, x_{2i} is a $1 \times k_2$ vector of additional instruments, and the equation for y_{2i} is written in reduced form. β and γ are vectors of parameters, and Π_1 and Π_2 are matrices of reduced-form parameters. We observe

Table 1. Descriptive statistics and correlations.

	Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12
1	IPO	0.01	0.09	1.00											
2	Number of VC firms	3.61	2.15	0.01	1.00										
3	Founding team size	1.86	1.04	0.03	0.08	1.00									
4	Geographic distance	1472	1452	0.01	0.11	0.03	1.00								
5	Lead VC reputation (Ln)	1.59	3.25	0.02	0.25	0.08	-0.09	1.00							
6	CVC relationships	0.32	0.47	-0.02	0.24	0.02	0.01	0.14	1.00						
7	Alliance relationships	0.08	0.27	0.01	0.05	-0.02	0.05	0.07	0.09	1.00					
8	New venture patent quality	36.3	242.9	0.00	0.01	0.16	-0.02	0.09	0.13	0.10	1.00				
9	Total investment amount (Ln)	0.83	3.31	0.02	0.34	0.04	0.05	0.31	0.25	0.14	0.11	1.00			
10	Prior industry experience	0.82	0.39	-0.03	0.03	0.18	-0.02	0.09	0.03	0.03	0.04	0.04	1.00		
11	Novice entrepreneurs	0.62	0.49	0.01	-0.01	-0.19	0.04	-0.02	0.03	0.01	-0.10	-0.01	-0.30	1.00	
12	Number of IPO firms	234.01	165.72	0.10	-0.09	0.01	-0.01	-0.09	-0.09	0.06	-0.04	-0.27	-0.03	0.03	1.00

$$y_{1i} = \begin{cases} 0y_{1i}^* < 0 \\ 1y_{1i}^* \geq 0 \end{cases}$$

We used *Founding team size* as an instrument to address the potential endogeneity regarding *Number of VC firms*. We measured *Founding team size* by counting the number of founders for a new venture. Research shows that founders' ties play an important role when new ventures seek external funding (e.g., Hallen 2008), thus a new venture with multiple founders may be able to have a large syndicate using the social relationships of the founders. However, the number of founders is unlikely to directly affect the likelihood of going public. Table 2 shows first-stage regression results for the endogenous variable, *Number of VC firms*. We find that *Founding team size* is positive and significant ($\beta = 0.16$, $p < 0.01$). Thus, in Models 2–6 of Table 3, we account for the endogeneity of number of VC firms.

In Table 3, Model 1 contains only the control variables. Model 2 introduces a linear term of *Number of VC firms*. Model 3 includes the number of VC firms as linear and quadratic terms to test H1. Model 4 introduces the main effect of *Geographic distance* and the interaction effect with number of VC firms squared to test H2. Model 5 introduces the main effect of *Lead VC reputation* and the interaction with the number of VC firms squared to test H3. Because a startup appears multiple times in our firm-year data structure, we clustered the standard errors by a new venture to address non-independence of observations (Williams 2000; Wooldridge 2002).

H1 posits an inverted-U-shaped relationship between the number of VC firms and the IPO likelihood. The results in Model 3 in Table 3 show that the linear term, number of VC firms, is positive and significant ($\beta = 1.434$, $p < 0.01$), and the number of VC firms squared is

Table 2. First-stage regression for VC syndicate size.

Variables	Model
Founding team size	0.16** (0.06)
CVC relationships	0.77** (0.16)
Alliance relationship	0.04 (0.27)
New venture innovation capability	−0.001** (0.00)
Total investment amount	0.19** (0.02)
Prior industry experience	−0.02 (0.24)
Novice entrepreneurs	0.01 (0.16)
Number of IPO firms	0.00 (0.00)
Constant	2.81** (0.39)
Industry indicators	Yes
Location indicators	Yes
Year indicator	Yes
Number of observations	1,112

The results are from the first-stage regression results in Model 2 of Table 3.

*** $p < 0.001$ ** $p < 0.01$, * $p < 0.05$ Robust standard errors clustered by startups in parentheses.

Table 3. Probit model accounting for syndicate size.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model6 (robustness)
Number of VC firms		0.433** (0.103)	1.434** (0.334)	2.205** (0.507)	1.928** (0.226)	1.736** (0.504)
Number of VC firms X Number of VC firms			-0.147** (0.035)	-0.257** (0.062)	-0.220** (0.025)	-0.199** (0.050)
Number of VC firms X Geographic distance in syndication				-0.001** (0.000)		
Number of VC firms^2 X Geographic distance in syndication				0.000** (0.000)		
Number of VC firms X Lead VC reputation					-0.284** (0.053)	-0.250** (0.082)
Number of VC firms^2 X Lead VC reputation					0.036** (0.007)	0.032** (0.009)
Geographic distance in syndication				0.001** (0.000)	-0.000 (0.000)	-0.000 (0.000)
Lead VC reputation				-0.007 (0.032)	0.373** (0.076)	0.302* (0.131)
CVC relationships	-0.320 (0.305)	-0.496** (0.167)	-0.236 (0.142)	-0.183 (0.143)	-0.245 (0.141)	-0.097 (0.122)
Alliance relationship	0.158 (0.435)	0.073 (0.269)	0.069 (0.227)	0.166 (0.227)	-0.016 (0.167)	0.056 (0.180)
New venture innovation capability	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Total investment amount	0.078 (0.054)	-0.042 (0.046)	-0.006 (0.036)	0.024 (0.029)	-0.006 (0.027)	0.004 (0.037)
Prior industry experience	-0.338 (0.289)	-0.205 (0.230)	-0.389* (0.171)	-0.180 (0.185)	-0.409** (0.156)	-0.295* (0.128)
Novice entrepreneurs	0.027 (0.268)	0.041 (0.156)	-0.015 (0.121)	0.085 (0.127)	-0.068 (0.098)	-0.038 (0.107)
Number of IPO firms	0.005** (0.001)	0.002 (0.002)	0.002 (0.002)	0.002 (0.001)	0.002 (0.001)	0.001 (0.001)
Constant	-3.203** (0.745)	-3.010** (0.730)	-3.620** (0.588)	-4.885** (0.386)	-3.606** (0.694)	-3.281** (0.285)
Industry indicators	Yes	Yes	Yes	Yes	Yes	Yes
Location indicators	Yes	Yes	Yes	Yes	Yes	Yes
Year indicator	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,112	1,112	1,112	1,112	1,112	1,112
Likelihood	-41.16	-2376.7	-1048.37	-530.63	-842.68	-946.08
AIC	110.32	4817.4	2164.73	1145.27	1769.36	1976.15

*** $p < 0.001$ ** $p < 0.01$, * $p < 0.05$.

Robust standard errors clustered by startups in parentheses.

negative and significant ($\beta = -0.147$, $p < 0.01$). Figure 4 shows the curvilinear relationship between the number of VC firms and the likelihood of IPO. As we expect, the relationship displays an inverted-U shape, and the inflection point of the inverted-U curve occurs at around 5. Thus, H1 is supported.

H2a and H2b present competing hypotheses regarding the moderating effect of geographic dispersion on the inverted-U relationship between the number of VC firms and the likelihood of IPO. H2a suggests that the more geographically dispersed the firms, the more pronounced the curve, whereas H2b suggested that the more geographically dispersed, the more flattened the curve. In Model 4, the coefficient on the interaction between geographic dispersion and the number of VC firms is negative and significant ($\beta = -0.001$, $p < 0.01$), whereas the coefficient on the interaction between geographic dispersion and the number of VC firm squared is positive and significant ($\beta = 0.0001$, $p < 0.01$). We draw Figure 5 to illustrate the differences in the shapes of the inverted-U

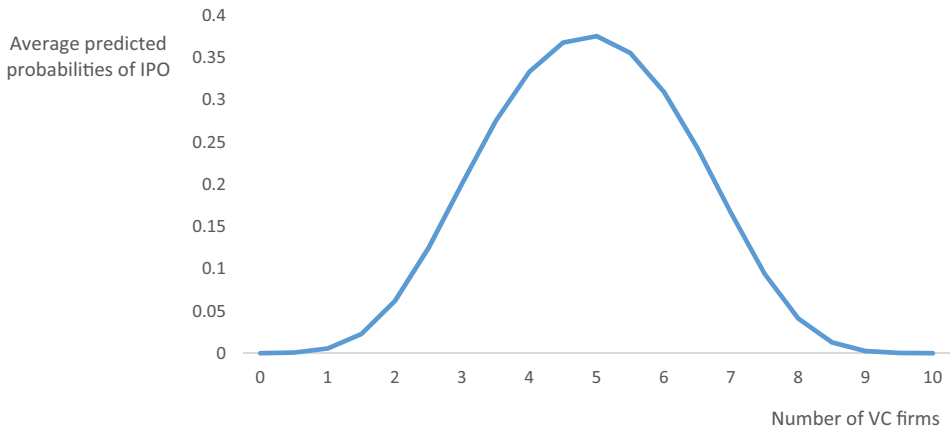


Figure 4. The relationship between syndicate size and IPO likelihood.

curve in different situations. Specifically, the curve indicating the low-geographic dispersion (25 percentile, solid line) has a more pronounced inverted-U shape than the one representing median situations. Conversely, when geographic dispersion among VC firms is very high (75 percentile, long-dashed line), the inverted-U shape curve between syndicate size and venture performance becomes flattened. Thus, H2b is supported.

H3 predicts that lead VC reputation would shift the inverse-U-shaped relationship between syndicate size and venture performance outward to the upper-right. In Model 5, the coefficient on the interaction between lead VC reputation and the number of VC firms is negative and significant ($\beta = -0.284$, $p < 0.01$), whereas the coefficient on the interaction between lead VC reputation and the number of VC firm squared is positive and significant ($\beta = 0.036$, $p < 0.01$). Figure 6 shows that when the lead VC is highly reputable (75 percentile, solid line), the inflection point occurs between 5 and 6 in terms of number of VC firms,

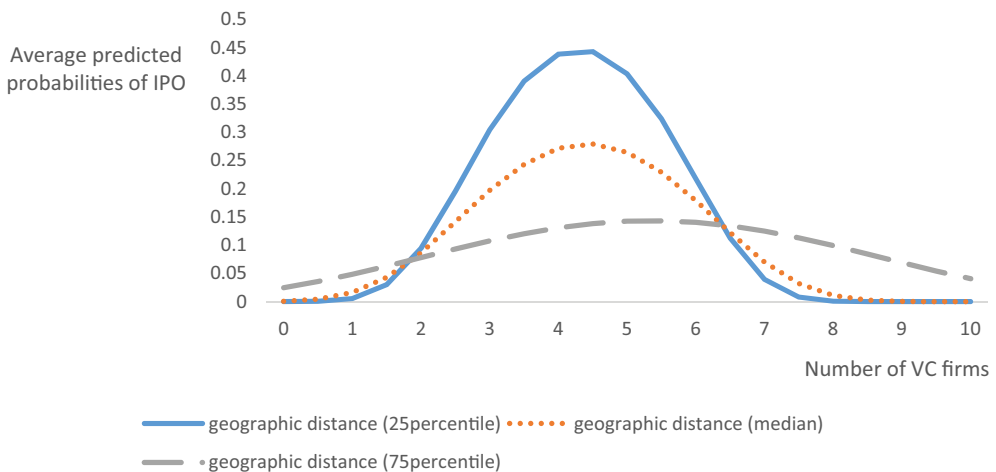


Figure 5. The moderating effect of geographic dispersion on the relationship between syndicate size and IPO likelihood.

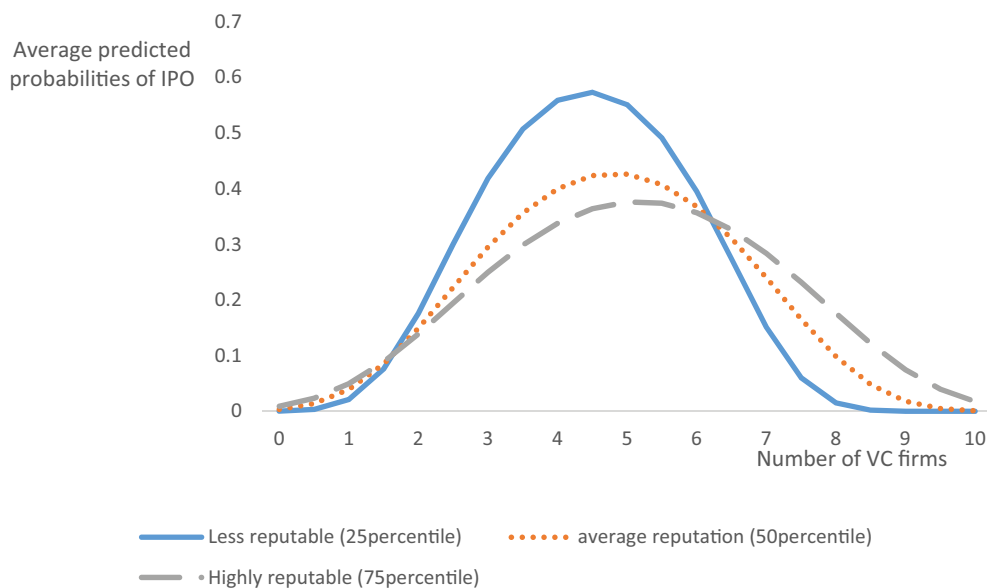


Figure 6. The moderating effect of lead VC reputation on the relationship between syndicate size and IPO likelihood.

suggesting that the syndicate can accommodate 5 or 6 VC firms without incurring as much coordination cost. This is larger than the syndicate size of when the lead VC's reputation is at 25 percentile or at the median. Intriguingly, however, the plot shows that the likelihood of IPO in a syndicate led by a highly reputable lead VC is lower at the inflection point than the one of the syndicates led by a less-reputable VC firm. Nonetheless, when there are more than 6 VC firms in a syndicate and when the lead VC is highly reputable (75 percentile), the likelihood of a venture funded by such a syndicate going public is highest with additional VC firms joining the syndicate, highlighting the benefit of a reputable lead VC in a particularly large syndicate. Thus, H3 is partially supported.

Robustness check and post-hoc analysis

We ran a number of tests to ensure the robustness of our results. First, IPO is typically considered as the most preferred outcomes for VC firms (Gompers and Lerner 2004). However, some view acquisition as another important outcomes for new ventures (e.g., Graebner and Eisenhardt 2004; Ma, Rhee, and Yang 2013). Thus, we ran a model using an alternative measure in which both acquisition exits and IPOs are coded as a measure of successful venture performance. As shown in Model 6 of Table 3, the results with this alternative measure are qualitatively similar to our main results in terms of the sign of the coefficients and plots.

Second, we operationalized the reputation of a lead VC firm by looking at the total number of its venture investees taken public prior to a given year. Other researchers have used a different measure for the reputation of a VC firm. Thus, we ran a model using the Lee-Pollock-Jin VC reputation index (Lee, Pollock, and Jin 2011; Pollock et al. 2015), and found that the results based on this alternative measure are consistent with our main results.

Third, our theory considers the outcome of VC syndicate as a net effect of resource heterogeneity and coordination cost, yet we are unable to measure each mechanism individually. To address this shortcoming, we performed some posthoc analyses to better understand the underlying mechanism. Specifically, we examine if partner firms have prior collaboration experience. If a newly joining VC firm had co-investment experience with other existing partners in the past, mutual trust and more stable role expectations based on repeated interaction are likely to mitigate the burden of coordination cost associated with a larger syndicate size (Gulati 1995; Ring and Van De Ven 1994). Thus, we created a measure, prior joint investment experience, by computing the average of the number of dyadic ties in the syndicate based on their co-investment experiences in the past five years. We ran a model using this measure and found that when the level of prior joint investment experiences in the syndicate is high, larger syndicate size is positively associated with higher likelihood of IPO without showing any negative effect in the range observed in our data. Although these findings are consistent with our speculation, we are cautious of interpreting this as conclusive because, in running the model, syndicates that have fewer than three partner firms were dropped from the analysis. Future study directly examining the dynamics of syndicate partner firms with a larger sample would be valuable.

Discussion and conclusion

We consider how the size of a VC syndicate affects venture performance. We find an inverse-U relationship between syndicate size and performance such that medium-sized syndicates perform better compared with small or overly large syndicates. We develop our theory based on the dynamics of VC firms in syndicates in providing resources and coordinating activities that influence the outcome of the venture investees. Although heterogeneous resources provided by a large syndicate can be beneficial to a new venture, exponentially increasing coordination costs outweigh such benefits beyond a certain size, leading to an inverted-U-shaped relationship between syndicate size and performance of the new venture the syndicate invests in. We also find that the relationship is moderated by two contingencies: geographical dispersion of syndication partners and reputation of lead VC firm. Specifically, geographic dispersion shifts the inverse U-curve downwards, suggesting that coordination costs due to geographical dispersion increase faster than benefits from resource heterogeneity as additional members join the syndicate. On the other hand, we find that syndicates led by highly reputable VC firms can accommodate a greater number of VC firms in the syndicate, supporting our argument that the presence of highly reputable lead VC mitigates the increase of coordination costs as syndicate size grows.

Contributions

Our study has important implications to several strands of literature. First, our study contributes to research on resource assembly of entrepreneurial ventures through VCs. The formation of a VC syndicate typically represents the first instance in which the separation of ownership and control arises in a new firm (Berle and Means 1932). When entrepreneurs receive funding from a VC syndicate and sell part of their venture equity stake to the VC partner firms, they share not only residual claims but also control rights

with those investors. In doing so, entrepreneurs may face a dilemma associated with the dual role that external investors play in resource provisioning and governance. Thus, achieving an optimal size of external stakeholders for resource reliance is particularly important for entrepreneurs. In this regard, this study offers entrepreneurs a more nuanced understanding on how the management of external constituents (i.e., the number of VC firms for the investment syndicate) affects the performance of new firms.

Second, our study contributes to research on VC firms. For VC firms, partnerships established through syndication networks are one of the most important strategies in achieving success (Hochberg, Ljungqvist, and Lu 2007; Hsu 2006). Numerous studies have documented that VC firms can be critical in the growth of new ventures by providing valuable financial and strategic resources through syndication (e.g., Brander, Amit, and Antweiler 2002). Although syndicate size is a fundamental determinant of syndication performance, prior studies have paid only secondary attention to it or controlled for it as an alternative explanation for other variables of interest. Moreover, empirical findings based on a handful studies have been mixed at best. Some researchers have found that larger syndicate size enhances the successful exit of portfolio firms (Giot and Schwienbacher 2007), whereas others have found that larger syndicate is detrimental to the exit and long-term performance of new ventures (Dimov and De Clercq 2006; Falconieri, Filatotchev, and Tastan 2019). Our study contributes to this literature by clarifying the influence of syndicate size. We examine both collaborative benefits and coordination challenges simultaneously, and provide a strong theoretical basis as well as empirical evidence. In this regard, our research also provides practical implications to VC firms. Specifically, our findings suggest that individual VC firms understand the performance implications of the size of the syndicate they are considering joining, as well as the characteristics of other investors such as geographic dispersion among partners. In addition, VC firms should consider the reputation of the lead VC firm in terms of its previous portfolio firms and their successful exit outcomes, in order to maximize the benefits of large syndicate while mitigating potential problems associated with the large size.

Third and more broadly, our study speaks to the literature on multiparty collaborations. Although scholars have examined how multiparty collaborations form and evolve (Li et al. 2012; Zhang, Gupta, and Hallen 2017), research on the outcome of multiparty collaboration is rather scarce. In a handful studies, some researchers have examined the benefits of multiparty alliances by focusing on the differential benefits that individual firms receive from participating in a multiparty alliance (e.g., Lavie, Lechner, and Singh 2007). Other researchers have investigated unplanned dissolutions of multiparty alliances by highlighting the downside of multiparty structures, such as the presence of cohesive subgroup and resulting divisive faultlines (Heidl et al., 2014). Our study contributes to this literature by showing that the performance of multiparty collaboration is a function of both resource benefits and coordination costs and identifying two important contingencies in the context of a VC syndicate.

Limitations and future research

As with other studies, our study is not without limitations that provide opportunities for future research. First, we find only partial support for the moderating effect of lead VC firm reputation on the relationship between syndicate size and venture performance.

Theoretically, reputable lead VC firms may increase the benefit of resource heterogeneity by inviting other firms that may possess more heterogeneous resources as the syndicate becomes larger. However, such firms could also help reduce coordination costs by influencing other firms to follow their leadership based on past experiences and knowledge in guiding ventures to successful exit outcomes. Although either of the two mechanisms could lead to the same result of shifting the curve outward, we are unable to disentangle the two effects empirically with our data. We unexpectedly find that the inflection point of a syndicate led by more reputable lead VC firms is lower than that of a syndicate led by less-reputable lead VC firms when the syndicate size is small, and such syndicates led by more reputable VC firms perform better only when the syndicate size is very large. These results appear to suggest that the decrease in coordination costs is the primary driver behind the moderating effect. However, this is only a speculation and future research that disentangles the two mechanisms would be valuable, perhaps through an in-depth qualitative study. Second, our results are based on taking on a sample of US-based ventures in the ICT sector. Although this sector has experienced one of the most vibrant VC investment activities and enabled us to control for industry heterogeneity, our findings may not be generalizable to other industries. Future research is needed to validate our findings in different contexts. Finally, we examined the performance implication of a multiparty collaboration effort in the context of the VC industry. Future studies could consider how partnership size affects performance in other contexts requiring collaborations by multiple actors. For instance, the number of co-authors in scientific projects has increased over time (Persson, Glänzel, and Danell 2004) at least in part due to the specialized knowledge required by a team. We suspect that the knowledge brought by diverse collaborators would be beneficial to the outcome of those projects, yet those teams could suffer greater coordination costs as their size increase. It would be interesting to find an optimal size in various contexts in addition to the VC syndication context.

Conclusion

As entrepreneurs assemble their resources from external sources to exploit an opportunity, they must carefully consider the dynamics of external resource providers to enhance their likelihood of success. As significant resource providers, venture capitalists play an important role in influencing entrepreneurial outcomes. We hope that our findings in this study can inform entrepreneurs on how to navigate such a process to enhance their performance.

Notes

1. <https://bothsidesofthetable.com/how-many-investors-are-too-many-b0ae2ccc523e>.
2. Specifically, we excluded the rounds at the following stages according to the VentureXpert database: acquisition, MBO, MBI, secondary buyout, pending acquisition, secondary purchase, open market purchase, PIPE, and real estate, as these investments typically do not constitute investments in high potential, yet high-risk startups.
3. Available at <http://bear.warrington.ufl.edu/ritter/ipodata.htm>.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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